ClipSAT : Cambridge & Edexcel Additional Mathematics 9606/4PM1 [ANSWER KEY]

Unit 5.01: Trigonometric Functions & Circular Measure

Q1. A circle sector has radius r = 6 text{ cm} and central angle theta = frac{pi}{3} text{ rad} . What is the arc length of the sector?

[x] A: frac{6pi}{3} text{ cm}
[ ] B: frac{36pi}{6} text{ cm}
[ ] C: 6pi text{ cm}
[ ] D: frac{6pi}{6} text{ cm}

Solution: Arc length formula in radians: s = rtheta = 6leftfrac{pi}{3}right = frac{6pi}{3} text{ cm} .

Q2. A circle sector has radius r = 8 text{ cm} and central angle theta = frac{pi}{4} text{ rad} . What is the arc length of the sector?

[x] A: frac{8pi}{4} text{ cm}
[ ] B: frac{64pi}{8} text{ cm}
[ ] C: 8pi text{ cm}
[ ] D: frac{8pi}{8} text{ cm}

Solution: Arc length formula in radians: s = rtheta = 8leftfrac{pi}{4}right = frac{8pi}{4} text{ cm} .

Q3. A circle sector has radius r = 10 text{ cm} and central angle theta = frac{pi}{5} text{ rad} . What is the arc length of the sector?

[x] A: frac{10pi}{5} text{ cm}
[ ] B: frac{100pi}{10} text{ cm}
[ ] C: 10pi text{ cm}
[ ] D: frac{10pi}{10} text{ cm}

Solution: Arc length formula in radians: s = rtheta = 10leftfrac{pi}{5}right = frac{10pi}{5} text{ cm} .

Q4. A circle sector has radius r = 12 text{ cm} and central angle theta = frac{pi}{6} text{ rad} . What is the arc length of the sector?

[x] A: frac{12pi}{6} text{ cm}
[ ] B: frac{144pi}{12} text{ cm}
[ ] C: 12pi text{ cm}
[ ] D: frac{12pi}{12} text{ cm}

Solution: Arc length formula in radians: s = rtheta = 12leftfrac{pi}{6}right = frac{12pi}{6} text{ cm} .

Q5. A circle sector has radius r = 14 text{ cm} and central angle theta = frac{pi}{7} text{ rad} . What is the arc length of the sector?

[ ] A: frac{196pi}{14} text{ cm}
[x] B: frac{14pi}{7} text{ cm}
[ ] C: 14pi text{ cm}
[ ] D: frac{14pi}{14} text{ cm}

Solution: Arc length formula in radians: s = rtheta = 14leftfrac{pi}{7}right = frac{14pi}{7} text{ cm} .

Q6. A circle sector has radius r = 16 text{ cm} and central angle theta = frac{pi}{8} text{ rad} . What is the arc length of the sector?

[x] A: frac{16pi}{8} text{ cm}
[ ] B: frac{256pi}{16} text{ cm}
[ ] C: 16pi text{ cm}
[ ] D: frac{16pi}{16} text{ cm}

Solution: Arc length formula in radians: s = rtheta = 16leftfrac{pi}{8}right = frac{16pi}{8} text{ cm} .

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Q7. A circle sector has radius r = 18 text{ cm} and central angle theta = frac{pi}{9} text{ rad} . What is the arc length of the sector?

[x] A: frac{18pi}{9} text{ cm}

[ ] B: frac{324pi}{18} text{ cm}

[ ] C: 18pi text{ cm}

[ ] D: frac{18pi}{18} text{ cm}

Solution: Arc length formula in radians: s = rtheta = 18leftfrac{pi}{9}right = frac{18pi}{9} text{ cm} .

Q8. A circle sector has radius r = 20 text{ cm} and central angle theta = frac{pi}{10} text{ rad} . What is the arc length of the sector?

[ ] A: 20pi text{ cm}

[ ] B: frac{400pi}{20} text{ cm}

[x] C: frac{20pi}{10} text{ cm}

[ ] D: frac{20pi}{20} text{ cm}

Solution: Arc length formula in radians: s = rtheta = 20leftfrac{pi}{10}right = frac{20pi}{10} text{ cm} .